# MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
*(A constituent unit of MAHE, Manipal)*

| DEPARTMENT OF CIVIL ENGINEERING | | | |
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| **END SEMESTER EXAMINATION** | | **NOVEMBER, 2024** | |
| **Level** | **B.Tech** | **Curriculum** | **2022** |
| **Program Name** | **Civil Engineering** | **Course Type** | **PE VII** |
| **Course Name** | **Remote Sensing and GIS** | **Course Code** | **CIE 4081** |
| **Semester** | **7th Semester** | **Max Marks** | **50** |
| **Date** | **20th November, 2024** | **Duration** | **3 Hours** |

**Instructions:**

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

| Q. No. | Question | Marks | NBA | | BL |
|---|---|---|---|---|---|
| | | | CO | PO | |
| **1A** | Explain the nearest neighbor sampling and bilinear interpolation procedures to correct for geometric distortions in remotely sensed images. | **4** | CO2 | 1,6,8,9,10 | **BL4** |
| **1B** | Explain linear contrast stretching method for image enhancement. | **3** | CO4 | 1, 4,5,6,7,8,9,10 | **BL4** |
| **1C** | Explain how GIS is a better database management tool than CAD/CAM systems and MS Excel. | **3** | CO3 | 1, 4,5,6,7,8,9,10 | **BL4** |
| **2A** | Explain any two unsupervised classification techniques. | **4** | CO2 | 1,6,8,9,10 | **BL4** |
| **2B** | Explain how remote sensing is used by geologists to explore new mineral deposits. | **3** | CO5 | 1, 4,5,6,7,8,9,10 | **BL4** |
| **2C** | Explain how is remote sensing used by policy makers for assessing agricultural productivity and estimating quantity of different crops grown in a region. | **3** | CO5 | 1, 4,5,6,7,8,9,10 | **BL4** |

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| **3A** | Explain any two use-cases of thematic maps generated using GIS in environmental and political science. | **4** | CO3 | 1, 4,5,6,7,8,9,10 | **BL4** |
| **3B** | Compare and contrast between the different kinds of scattering experienced by electromagnetic waves as they pass through Earth's atmosphere. | **3** | CO1 | 1,6,8,9,10 | **BL5** |
| **3C** | Explain spectral ratioing. Explain its advantages and disadvantages. | **3** | CO4 | 1, 4,5,6,7,8,9,10 | **BL4** |
| **4A** | "Sun is the ideal blackbody for electromagnetic remote sensing in the visible range according to Boltzmann's law and Wein's displacement law." Defend this statement with support from a neat diagram. | **5** | CO1 | 1,6,8,9,10 | **BL5** |
| **4B** | Explain the key components of GIS. | **3** | CO3 | 1, 4,5,6,7,8,9,10 | **BL4** |
| **4C** | Explain what is grey level thresholding. | **2** | CO4 | 1, 4,5,6,7,8,9,10 | **BL4** |
| **5A** | Explain what is a histogram. Show histogram characteristics for the following types of scenes with a neat diagram:<br>a) low radiance scene<br>b) high radiance scene<br>c) low contrast scene<br>d) high contrast scene | **5** | CO4 | 1, 4,5,6,7,8,9,10 | **BL4** |
| **5B** | Explain any 6 advantages of GIS. | **3** | CO3 | 1, 4,5,6,7,8,9,10 | **BL4** |
| **5C** | Explain high-pass filtering with an example. | **2** | CO4 | 1, 4,5,6,7,8,9,10 | **BL4** |

## MARK DISTRIBUTION (CO-wise & BTL-wise)

| CO / CLO | Total Marks |
|---|---|
| **1** | 8 |
| **2** | 8 |
| **3** | 13 |
| **4** | 15 |
| **5** | 6 |

| BL | Total Marks |
|---|---|
| **2** | **0** |
| **3** | **0** |
| **4** | **42** |
| **5** | **8** |